

motion - practice problems

name _____

1. A small camera drone starts 2 m above the ground while filming a school field. It rises to 14 m above the ground, and the camera records for 9 s. What is its vertical displacement?
2. A student starts at a locker marked -12 m and walks to a classroom marked 35 m. What is the student's displacement?
3. A runner sprints 0.10 km east, turns around, and jogs 35 m west. What are the runner's distance traveled and displacement?
4. A city bus leaves the science museum stop at 7:48 AM and follows a 6.2 km route to school, arriving at 8:13 AM. How long was the ride in minutes?

5. A runner jogs for 12 s and covers 80 m on the school's running track. What is the runner's displacement?
6. A skateboarder goes from the 0.008 km mark to the 68 m mark in 15 s. What is the skateboarder's average velocity?
7. A dog runs 18 m east, then 6 m west, in 8 s. What are the dog's distance traveled and average velocity?
8. An elevator rises with an average velocity of 1.5 m/s. How long does it take to rise 36 m?
9. A train backs up slowly at 5.4 km/h for 20 s while coupling to another car. Let forward be positive. What is the train's displacement?

10. A position vs. time graph goes through the points (2 s, 5 m) and (8 s, 29 m). What is the slope of the graph?
11. A position vs. time graph falls from 40 m at 0 s to -20 m at 10 s. What average velocity does the graph show?
12. A skateboarder speeds up from 1.5 m/s to 7.5 m/s in 4 s. What is the skateboarder's average acceleration?
13. A bike is moving forward at 9 m/s and comes to rest in 3 s. What is the bike's average acceleration?
14. A cart is moving left at 4 m/s, then 5 s later it is moving right at 6 m/s. Let right be positive. What is the cart's average acceleration?

15. A velocity vs. time graph goes through the points (3 s, 2 m/s) and (11 s, -6 m/s). What is the slope of the graph?

16. A cyclist on a straight bike path starts a timer at the 0.020 km mark and is at the 170 m mark after 0.50 min. During that same time, the bike computer reports that the cyclist's velocity changes from 2 m/s to 8 m/s. Find the displacement, average velocity, and average acceleration.

17. Two cars drive toward each other on a straight east-west road. Let east be positive. At the same instant, one car is at the 0 m marker and travels east at a constant 20 m/s. The other car is at the 600 m marker and travels west at a constant 10 m/s. How long until the cars meet?